INCH-POUND

MIL-PRF-8805/3J
3 September 1999
SUPERSEDING
MIL-S-8805/3H
30 November 1987

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSH, 10 AMPERES OR LOW LEVEL, DUSTTIGHT

This specification forms a part of MIL-PRF-8805, dated 23 January 1998, and is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.

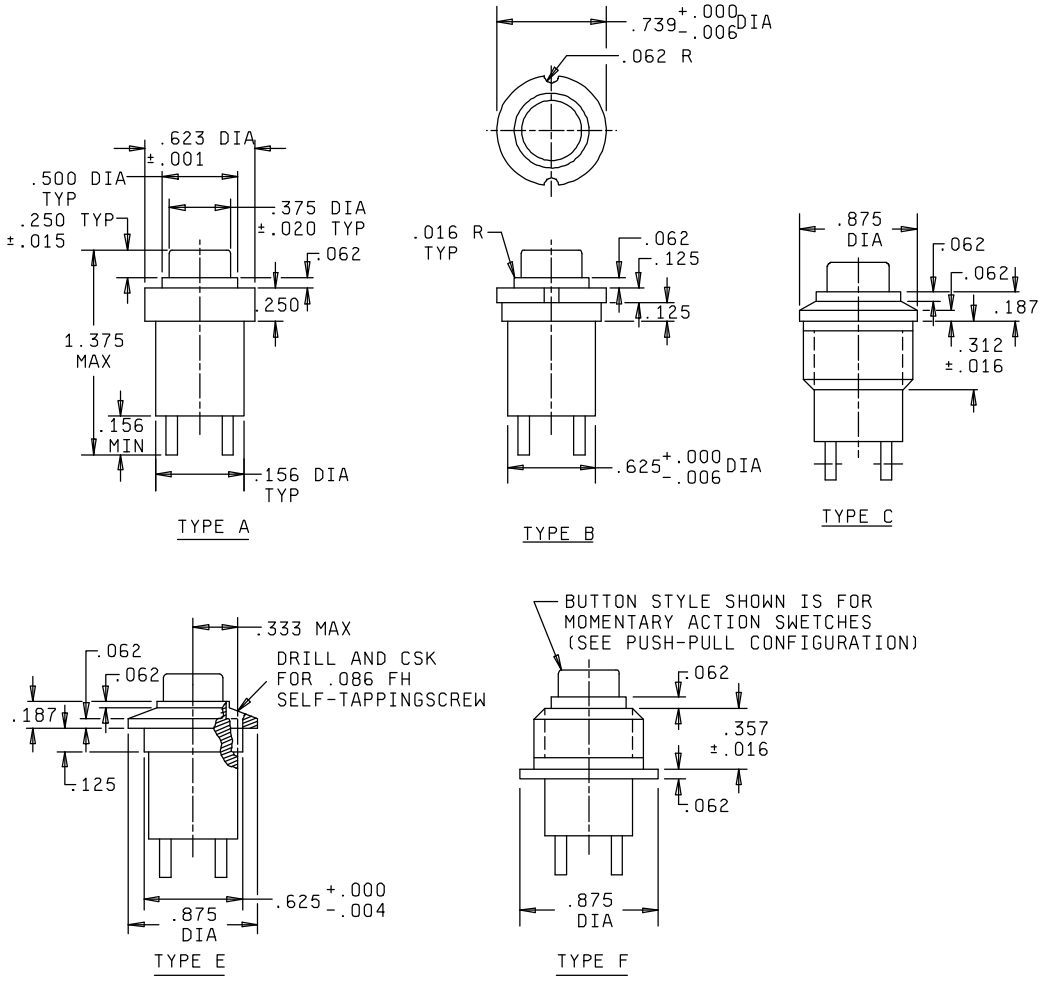
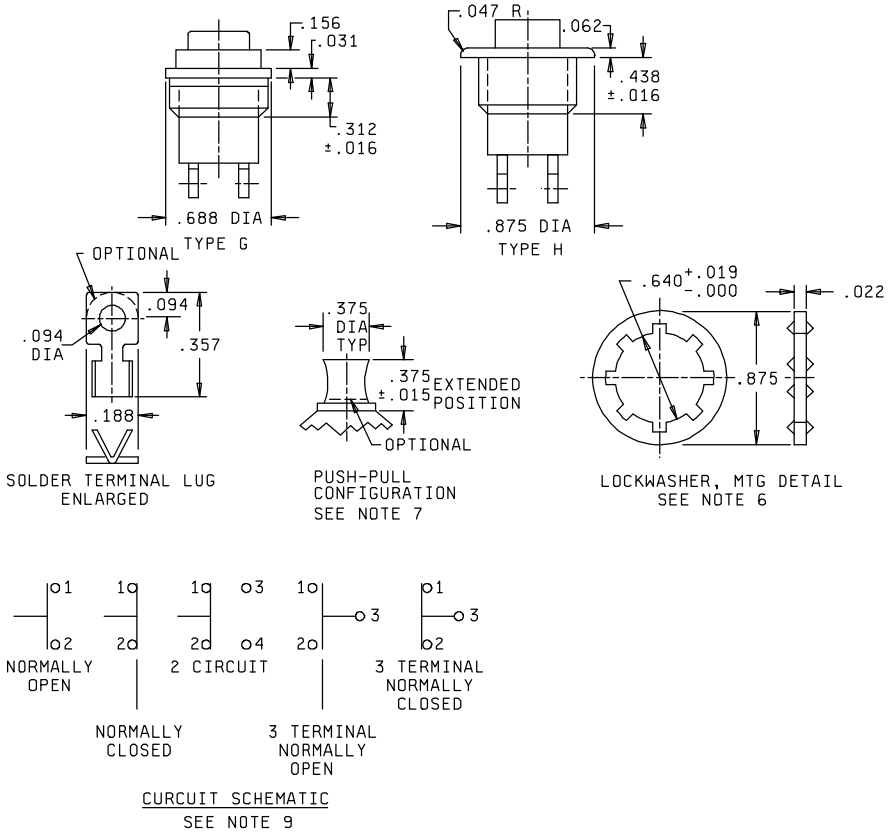


FIGURE 1. Configuration and dimensions.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.250	6.35	.623	15.82
.004	0.10	.047	1.19	.312	7.92	.625	15.88
.006	0.15	.062	1.57	.333	8.46	.640	16.26
.015	0.38	.094	2.39	.375	9.53	.688	17.48
.016	0.41	.125	3.18	.438	11.13	.739	18.77
.019	0.48	.156	3.96	.500	12.70	.875	22.23
.020	0.51	.187	4.75	.562	14.27	1.375	34.93
.022	0.56	.188	4.78				

NOTES:

1. Dimensions are in inches.
2. Unless otherwise stated, tolerance is ±.010 (0.25 mm).
3. Mounting thread .625-24 UNEF-2A applies to type C, F, G, and H. All threads to be within .10 2.54 mm) of shoulder.
4. All terminal threads are .086-56UNC-28.
5. Hardware for each terminal includes one screw, one terminal lug, and one lockwasher.
6. Types C, F, G, and H shall be furnished with suitable mounting hardware.
7. Overall height of switch with push-pull activator is 1.530 (38.66 mm) maximum in extended position.
8. Metric equivalents are given for general information only.
9. For push-pull configuration, circuit schematic is shown with actuator in extended position.

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Enclosure design: 2 (dust-tight).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: H for monetary type only (no contact opening or closing in excess of 20 milliseconds). M for other units.

Sinusoidal vibration grade: 2 (10 to 2,000 Hz).

Weight: .050 pound maximum.

Solderability: Applicable to screw tabs and solder lugs.

Operating characteristics:

Total plunger travel: .085 ± .015 inch.

Operating force:

Momentary: 4 ±1 pounds.

Push-pull: 0.5 to 2 pounds.

Releasing force: 1 pound minimum, monetary.

Terminal strength torque test: Not applicable.

Strength of actuator and stop: 75 pounds for 3 terminals circuitry; 40 pounds for all others.

Finish: The switch case shall be corrosion resistant , lusterless black.

Dielectric withstanding voltage:

Sea level: 1,000 V rms.

After electrical endurance: 200 V rms.

Intermediate current: 1/

MS25089-1, MS25089-2, and MS25089-3: 200,000 cycles.

MS25089-4 and MS25089-5: 50, 000 cycles.

1/ Prior to test, units preconditioned at 85°C for 48 hours.

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Mechanical endurance (10 to 15 cpm):

MS25089-1, MS25089-2, and MS25089-3:

At room temperature: 200,000 cycles.

At -55°C and +85°C: 100,000 cycles each.

MS25089-4 and MS25089-5: 50,000 cycles.

Electrical endurance: 25,000 cycles at sea level; 5,000 cycles at 50,000 feet.

Electrical ratings:

Central terminal ratings: 3 amperes maximum.

Other than center terminals: See table I.

TABLE I. Electrical ratings, power circuit switches. 1/

Load	Sea level		50,000 feet
	28 V dc	115/V ac, 60/400 Hz	28 V dc
	Amperes	Amperes	Amperes
Resistive ----------	10	10	- - -
Inductive ----------	5	5	5
Lamp----------	3	3	- - -

1/ Center terminal rating: 3 amperes maximum.
2/ The 400 Hz ratings are shown for information only and do not require electrical endurance testing.

Sand and dust: Applicable.

Electrical logic circuit (5.0 V dc, .010 ampere): Applicable when specified (see table II): 25,000 cycles at an actuation rate of 120 cycles per minute maximum with no "stick", or "misses", allowed when tested in accordance with ANSI/EIA RS448, method 17 as follows:

Test condition:

Each switch contact shall be tested using a 5.0 ±0.5 V dc, 10 ±1 mA resistive load. During each closure of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact static closure. The switch contacts need not be monitored until 10 milliseconds after the initial contact closure to exclude any contact bounce. During each opening of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less thn 50 percent of each contact opening.

A voltage of 2.1 volts or greater across the switch terminals shall constitute a contact "miss" (failure to properly close the circuit). A voltage drop of less than 90 percent of the open-circuit voltage shall constitute a contact "stick" (failure to properly open the circuit).

The monitoring device shall either record the number of contact closures at which "sticks" and/or "misses" occur, or discontinue the test when "sticks" and/or "misses" occur.

Operating temperature:

a. 25 percent of the test cycle at -55°C specified.

b. 25 percent of the test cycle at room ambient conditions.

c. 50 percent of the test cycle at +85°C.

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Qualification:

Group submission: See table III.

Group A inspection: When low level circuit applies, switches shall not be subjected to loads greater than 100 milliamperes at an open circuit voltage of 6 volts maximum dc or peak ac.

Part number: See table II.

TABLE II. Part numbers and characteristics.

Type	Momentary action switches			Push-pull switches			Switch circuit
	Button color			Button color			
	Black	Red	White	Black	Red	White	
A	1A 2A 3A 4A 5A	1AR 2AR 3AR 4AR 5AR	1AW 2AW 3AW 4AW 5AW	None			NO NC 2 circuit 3 term NO 3 term NC
B	1B 2B 3B 4B 5B	1BR 2BR 3BR 4BR 5BR	1BW 2BW 3BW 4BW 5BW	None			NO NC 2 circuit 3 term NO 3 term NC
C	1C 2C 3C 4C 5C	1CR 2CR 3CR 4CR 5CR	1CW 2CW 3CW 4CW 5CW	1C-1 2C-1 3C-1 4C-1 5C-1	1CR-1 2CR-1 3CR-1 4CR-1 5CR-1	1CW-1 2CW-1 3CW-1 4CW-1 5CW-1	NO NC 2 circuit 3 term NO 3 term NC
E	1E 2E 3E 4E 5E	1ER 2ER 3ER 4ER 5ER	1EW 2EW 3EW 4EW 5EW	None			NO NC 2 circuit 3 term NO 3 term NC
F	1FR 2FR 3FR 4FR 5FR	1FW 2FW 3FW 4FW 5FW	1F 2F 3F 4F 5F	1F-1 2F-1 3F-1 4F-1 5F-1	1FR-1 2FR-1 3FR-1 4FR-1 5FR-1	1FW-1 2FW-1 3FW-1 4FW-1 5FW-1	NO NC 2 circuit 3 term NO 3 term NC
G	1GR 2GR 3GR 4GR 5GR	1GW 2GW 3GW 4GW 5GW	1G 2G 3G 4G 5G	1G-1 2G-1 3G-1 4G-1 5G-1	1GR-1 2GR-1 3GR-1 4GR-1 5GR-1	1GW-1 2GW-1 3GW-1 4GW-1 5GW-1	NO NC 2 circuit 3 term NO 3 term NC
H	1HR 2HR 3HR 4HR 5HR	1HW 2HW 3HW 4HW 5HW	1H 2H 3H 4H 5H	1H-1 2H-1 3H-1 4H-1 5H-1	1HR-1 2HR-1 3HR-1 4HR-1 5HR-1	1HW-1 2HW-1 3HW-1 4HW-1 5HW-1	NO NC 2 circuit 3 term NO 3 term NC

1/ Low level circuit and electronic logic circuit does not apply for dash numbers shown. To specify switches with low circuit and electronic logic circuit applicable, add the letter "L" to the dash number.

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Examples of part numbers:

MS25089-4C = Switch, Push, Dusttight, 3 Terminal (NO), Type C, Momentary, Black Button.

MS25089-4CR-1 = Switch, Push, Dusttight, 3 Terminal (NO), Type C, Push-Pull, Red Button.

MS25089-4CL = As above, except low level circuit and electronic logic circuit applies. Power circuit electrical ratings do not apply.

MS25089-4CR-1L = As above, except low level circuit and electronic logic circuit applies. Power circuit electrical ratings do not apply.

TABLE III. Qualification inspection.

Examination or test	Basic switch	Other switch samples	Extent of approval	
Qualification 1/ inspection table of MIL-PRF-8805	Required sample units of any 2 circuit configurations 2/	Required number of sample units of 3 terminal NO or 3 terminal NC (groups VII and x tests, qualification inspection table, MIL-PRF-8805 plus 2 additional sample units shall be subjected to electronic logic circuit test and operating characteristics) 2/	All	In addition, 2 sample units of each of the other configurations shown on this specification sheet shall be submitted. Visual and mechanical examination.

1/ Electrical endurance tests shall be performed at 28 V dc and 115 V ac, 60 Hz only.
2/ One-half of sample units shall be momentary and one-half shall be push-pull.

Custodians:
Air Force - 11
Army - CR
DLA - CC
Navy - EC

Review activities:
Air Force - 19, 99
Army - AR, AV, MI
Navy - AS, OS, MC, SH

Preparing activity:
DLA - CC

(Project 5930-1691-01)

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